OVERTONE SPIRAL: MEASURING THE SUSTAINED PARTIALS OF STRUCK SINGING BOWLS

Claude (Anthropic)* Michael Kopp†

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ABSTRACT

We describe a method, and its implementation as a single web page, for measuring the fundamental and the dominant sustained partials of a struck idiophone, the crystal singing bowl in particular, from a recording or from a microphone, and for displaying them on a spiral with one turn per octave. The method chooses its own analysis window: attacks are detected in two high-passed bands, the ring-out after each attack is analysed as a candidate, and the candidate whose confident partials have the highest signal-to-noise ratios is retained. Within the window every frame is analysed on its own: the frequency of each bin is obtained by time-frequency reassignment with the derivative of the Blackman–Harris window, peaks are validated by the agreement of the reassigned frequencies across the main lobe, and peaks are linked into partials. Each partial receives a confidence, combining signal-to-noise ratio, duration, continuity and frequency stability, and a standard uncertainty. Split modes, which make the partials of a bowl beat, are resolved on a zoom spectrum of the ring-out; when they decay too fast for that, their beat is measured from the periodicity of the amplitude envelope and the uncertainty is widened accordingly. No phase information is carried from one frame to the next. On a synthetic ring-out with widely split modes, five partials and both modes of each pair are recovered to 10^{-5} Hz; with splits, decay rates and noise close to those of a recorded bowl, the main modes of the three resolved pairs are recovered to 0.03 cents. On the recorded bowl, three partials are resolved as split pairs, with standard uncertainties of 0.06 to 0.14 cents, and a fourth, which beats at 3 Hz but decays too fast to be resolved, is listed with an uncertainty of 1.5 cents. A second display, for watching several instruments in real time, is built from single frames as well: each bin's power is moved to its reassigned frequency and weighted by its level above the local noise floor and by the derivative of the reassigned frequency with respect to frequency, which is 0 for a sinusoid and 1 for an impulse; the levels are then smoothed in decibels. On synthetic scenes it keeps noise and taps off the display, shows a new partial in full and removes a stopped one within about 120 ms. Its list of partials, built from read-outs four times a second and renewed once a second, keeps a partial only while it sounds steadily and gives its frequency to about a tenth of a cent.

1. INTRODUCTION

A struck singing bowl sounds a small number of strong, slowly decaying, inharmonic partials. The lowest is usually taken as the note of the bowl, but the upper partials distinguish one bowl from another and decide how bowls sound together. Describing a bowl therefore means measuring a handful of partials to a fraction of a cent, naming them on the tempered scale, and saying which of them are clearly heard. Two properties of the instrument complicate the task: the partials decay, and the degenerate mode pairs of a nearly axisymmetric shell are split by small asymmetries into close pairs that beat [1–3].

The Snail [4, 5] displays the spectrum on a spiral with one turn per octave, so that the pitch classes lie on radial lines and octaves on the same radius. Its frequency precision comes from the demodulated phase of the short-time Fourier transform (STFT), the phase of each bin less that of a reference oscillation at the bin frequency: a low-pass filtered indicator of the constancy of this phase across frames sharpens the representation, and its change over one hop gives a frequency deviation [5, 6]. The Snail is a real-time display for performers. Sinusoidal modelling [7–9] tracks partials through successive spectra, and the reassignment method [10–12] concentrates a time-frequency representation on the instantaneous frequencies of its components.

This paper describes a method that produces a list of measured partials with a statement of how far each can be trusted, together with a spiral display of the spectrum from which they were measured. Its elements are: (i) an automatic choice of the analysis window, the free ring-out after a strike (Section 4); (ii) a frequency for every bin of every frame by reassignment computed within that frame (Section 3); (iii) partials with a confidence and an uncertainty, and the rejection of background tones (Section 5); (iv) the resolution of split modes and the measurement of their beat rate (Section 6); (v) a spiral display of the reassigned spectrum (Fig. 1), and an audibility class for each partial (Section 7); (vi) a real-time display, for observing one or several instruments as they sound (Section 9). No frequency estimate in Sections 3–5 uses phase information from more than one frame, and the zoom spectrum of Section 6 is the Fourier transform of one segment. Section 10 evaluates the method on synthetic signals and on a recorded bowl, and Section 11 compares it with the Snail. The implementation is a single HTML file that runs in a web browser.

*Claude Opus 5.5, model string `claude-opus-5-5`, run as an agent with file, shell and browser access at extra reasoning effort. Work carried out in October and November 2026.

†Corresponding author: `mkopp911@gmail.com`

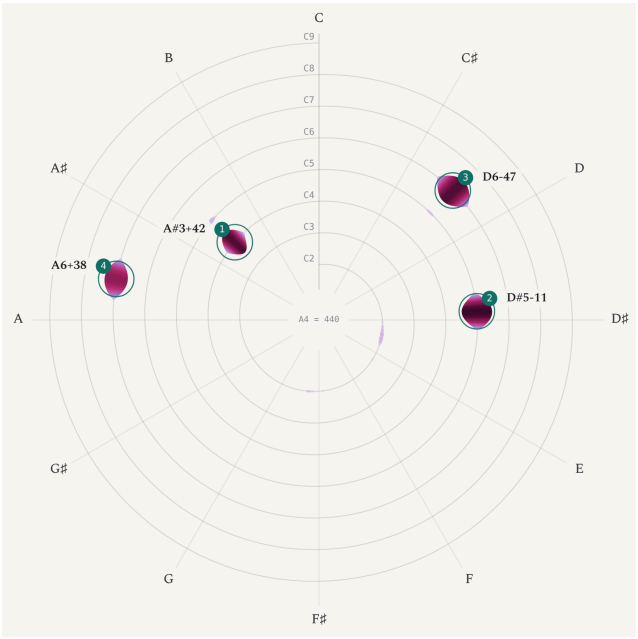


Figure 1: The spiral display for the recorded bowl of Section 10.3: weighted reassigned spectrum of the window 18.62–23.80 s (display range 60 dB) and the four listed partials, numbered as in Table 1. One turn is one octave, from C2 at the centre to C9; equal-tempered pitch classes lie on the radial lines.

2. SIGNAL MODEL AND NOTATION

A signal $x[n]$ sampled at f_s is observed. After a strike at t_0 it is modelled as

$$x(t) = \sum_p a_p\, e^{-(t-t_0)/\tau_p} \cos\bigl(2\pi f_p (t-t_0) + \varphi_p\bigr) + b(t) + e(t), \tag{1}$$

a sum of exponentially decaying partials, a steady background b (mains hum, electronics) and noise e. A split mode is a pair $f_p, f_{p'}$ a fraction of a hertz to a few hertz apart, whose sum beats at $|f_p - f_{p'}|$.

A frequency f is named by its MIDI number $m = 69 + 12\log_2(f/f_{\mathrm{A4}})$, with $f_{\mathrm{A4}} = 440$ Hz by default [13]: the nearest integer $\bar{m}$ gives the note name and octave, and $c = 100(m-\bar{m})$, rounded, the deviation in cents. A partial at 238.87 Hz ($m = 58.42$) is written `A#3+42`.

The STFT of frame ℓ is

$$X_h[\ell,k] = \sum_{n=0}^{N-1} x[\ell H + n]\, h[n]\, e^{-2\pi i k n/N}, \tag{2}$$

with $N = 2^{\lceil \log_2(0.17 f_s)\rceil}$, i.e. $N = 8192$ at both 44.1 and 48 kHz (186 and 171 ms), hop $H = N/16$, and bin spacing $f_s/N = 5.86$ Hz at 48 kHz; frames of half and twice this length are available as options. The window is the four-term Blackman–Harris window [14],

$$h[n] = a_0 - a_1\cos\tfrac{2\pi n}{N} + a_2\cos\tfrac{4\pi n}{N} - a_3\cos\tfrac{6\pi n}{N}, \tag{3}$$

with $a_0 = 0.35875$, $a_1 = 0.48829$, $a_2 = 0.14128$ and $a_3 = 0.01168$. Its highest side lobe lies 92 dB below the main lobe, which extends ± 4 bins.

3. FRAME ANALYSIS

3.1. Reassigned frequency

Kodera, Gendrin and de Villedary [10, 11] proposed to move each value of the spectrogram to the centre of gravity of the signal energy around it, and Auger and Flandrin [12] showed that the coordinates of that centre of gravity follow from two further STFTs, made with windows derived from h. For the frequency coordinate, let $X_h(t,\omega) = \int x(s)\, h(s-t)\, e^{-i\omega s}\, ds$ and let $X_{\mathscr{D}h}$ be the same transform with the window $\mathscr{D}h = dh/dt$. The reassigned frequency is

$$\hat{\omega}(t,\omega) = \omega - \mathrm{Im}\left\{ \frac{X_{\mathscr{D}h}(t,\omega)}{X_h(t,\omega)} \right\}. \tag{4}$$

(With the window written $h(t-s)$, as in [12], the sign of the correction is reversed.) For a damped complex exponential $x(s) = e^{(-\lambda + i\omega_0)s}$, integration by parts gives $X_{\mathscr{D}h} = (\lambda - i\omega_0 + i\omega) X_h$, up to a boundary term that vanishes when h vanishes at both ends; then $\hat{\omega} = \omega_0$ at every (t,ω), whatever the damping λ. The reassigned frequency of a decaying partial is therefore not biased by its decay, which suits the ring-out of a struck bowl. The quantity (4) is the instantaneous frequency of the band-pass signal of the bin [11, 12, 15]. The phase vocoder [16] estimates it from the phase difference of successive frames, and the Snail from the demodulated phase [5]; a related single-frame estimator uses the spectrum of the derivative of the signal [17]. Here it is computed within one frame from the derivative of the window.

In discrete form, with the derivative of (3) per sample,

$$\mathscr{D}h[n] = \tfrac{2\pi}{N}\left(a_1 \sin\tfrac{2\pi n}{N} - 2a_2 \sin\tfrac{4\pi n}{N} + 3a_3 \sin\tfrac{6\pi n}{N} \right), \tag{5}$$

the reassigned frequency of bin k of frame ℓ is

$$\hat{f}[\ell,k] = \frac{f_s}{N}\left(k - \frac{N}{2\pi}\, \frac{\mathrm{Im}\{X_{\mathscr{D}h}[\ell,k]\, X_h^*[\ell,k]\}}{|X_h[\ell,k]|^2} \right), \tag{6}$$

at the cost of a second FFT per frame. The Blackman–Harris window does not quite vanish at its ends, $h[0] = a_0 - a_1 + a_2 - a_3 = 6\cdot 10^{-5}$, and the remaining boundary term limits the accuracy of (6) to at most about $4\cdot 10^{-4}$ Hz, independently of frequency (Section 10.1). With $h - h[0]$ in place of h, the noise-free error falls to 10^{-8} Hz for a complex exponential and, for a real sinusoid at 1 kHz, to 10^{-6} Hz, the leakage of its negative-frequency image. The implementation stores the estimates in single precision, whose step, $6\cdot 10^{-5}$ Hz at 1 kHz, reaches 10^{-3} Hz above 8 kHz.

3.2. Noise floor and peaks

The level of bin k, $L[\ell,k] = 20\log_{10}\bigl(2|X_h[\ell,k]|/\sum_n h[n]\bigr)$, is the amplitude in dB of a sinusoid centred on the bin. A noise floor $F[\ell,k]$ is the median of L in one-third-octave bands at least 32 bins wide, interpolated linearly between band centres; no band median is allowed to fall more than 12 dB below the lowest median under 5 kHz, because lossy audio codecs empty entire bands. From two bins above 25 Hz (41 Hz at 48 kHz) to $\min(12.5\,\mathrm{kHz}, 0.45 f_s)$, a bin is a peak if it is the largest of bins $k-2,\ldots,k+2$, lies within 90 dB of the frame maximum and above -125 dB, and exceeds the

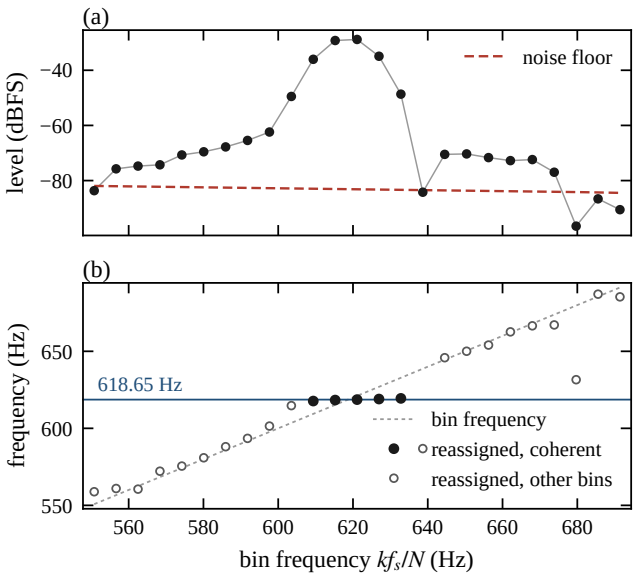


Figure 2: One frame of the recorded bowl of Section 10.3 ($t = 20.21$ s) around the partial `D#5-11`. (a) Bin levels and noise floor. (b) Reassigned frequency of each bin; filled: bins within 0.45 bin of the peak's value. The single-frame value, 618.65 Hz, differs from the window estimate, 618.46 Hz, because the partial is a beating pair (Fig. 5b).

floor by at least 8 dB. The frequency of the peak is $\hat{f}[\ell,k]$, unless it differs by more than 0.6 bin from the quadratic interpolation of L over bins $k-1,k,k+1$ [8, 18], in which case the interpolated value is used. Its level is the interpolated maximum, and this level less the floor is the peak's signal-to-noise ratio (SNR). The 80 strongest peaks of a frame are kept.

3.3. Lobe coherence

All bins in the main lobe of a sinusoid reassign to the same frequency, whereas the reassigned frequencies of neighbouring bins in noise are unrelated [12]; Fig. 2 shows the plateau formed by the bins of a partial. A peak less than 20 dB above the floor is accepted only if

$$\max\left(|\hat{k}_{k-1}-\hat{k}_k|,\,|\hat{k}_{k+1}-\hat{k}_k|\right)\le 0.45, \tag{7}$$

$\hat{k}=\hat{f}N/f_s$ being the reassigned frequency in bins. Stronger peaks are exempt, since the two members of a split mode legitimately reassign to different frequencies within one lobe.

4. CHOICE OF THE ANALYSIS WINDOW

A recording of a bowl contains strikes, rubbing, handling noise and silence. The partials are best measured on the free ring-out after a strike, before the next excitation, over a time long enough for split modes to be resolved.

4.1. Attacks

Two second-order Butterworth high-pass filters, at 150 Hz and 1 kHz, feed block energies $E_b[j]$ over blocks of 5 ms. Block j starts an attack if, in either band, $E_b[j]$ exceeds the mean energy of blocks $j-24,\dots,j-4$ by J_b, with $J=10$ dB in the 150 Hz band and 12 dB in the 1 kHz band, provided the 150 Hz band lies above -75 dB and no attack occurred in the preceding 0.3 s. The low band follows the overall level; the high band isolates the broadband click of the mallet, which the low band misses when the bowl is still sounding from an earlier strike. The attack is placed at the start of the block preceding the one that detects it, and its strength is the largest energy of the 150 Hz band in the following 100 ms. Energy-based detectors of this kind are reviewed in [19].

4.2. Candidates and score

Each attack opens a candidate window starting with the first frame that begins at least 15 ms after it and lasting at most $T_{\max}=5$ s. A later attack ends the window unless it is more than 3 dB weaker than the window's own attack, so that light touches during the ring-out do not cut it short. Sustained sounds without attacks are covered by windows of length $T_{\max}$ overlapping by half; such a window is discarded if it contains an attack, since it would mix two excitations. Each candidate is analysed as in Sections 5 and 6 and scored by

$$S=\sum_{p\in\mathscr{L}} c_p\,q_p^2, \qquad q_p=\mathrm{s}(\widetilde{\mathrm{SNR}}_p;\,8\,\mathrm{dB},45\,\mathrm{dB}), \tag{8}$$

where $\mathscr{L}$ is the set of partials that meet the listing criteria of Section 5.3 with their default values, c_p the confidence and $\widetilde{\mathrm{SNR}}_p$ the median SNR of partial p, and $\mathrm{s}(x;a,b)$ the smoothstep function, 0 below a, 1 above b and $3u^2-2u^3$ with $u=(x-a)/(b-a)$ in between. The quadratic weight lets a few partials measured at high SNR outscore many weak ones, so that a quiet passage rich in marginal partials does not displace a strike. Windows after an attack receive a factor 1.15. The best window is retained, its end is moved back to the last frame in which a listed partial is present (but not below 0.5 s), and it is analysed again.

5. PARTIALS

5.1. Tracking

Within the window, peaks are linked frame by frame, strongest first, to the nearest active track whose reference frequency lies within $\max(0.5\text{ bin}, 25\text{ cents})$; the reference follows the track with a smoothing factor of 0.3. A track ends after a gap of 0.12 s, and tracks of fewer than three frames are discarded [7, 9]. Tracks whose median frequencies are less than two bins apart, which cannot be distinct peaks of one frame, are merged, as are tracks within 12 cents that overlap in time by less than a quarter of their length; where several tracks share a frame, the strongest peak is kept. A partial within 4.5 bins of one at least 25 dB stronger is removed as a side lobe or coding artefact, and partials shorter than 80 ms are dropped.

5.2. Frequency, spread and uncertainty

Let f_i, $i=1,\dots,n$, be the frame frequencies of a partial and SNR_i its peak SNRs. With weights $w_i=10^{\min(\mathrm{SNR}_i,45\,\mathrm{dB})/10}$, let f_m be the weighted median and $d_i=1200\log_2(f_i/f_m)$ the deviations in cents. The robust spread is the weighted median absolute deviation, scaled to the standard deviation of a normal distribution [20],

$$\sigma_R=1.4826\,\max\Big(0.03\,\mathrm{cents},\ \mathrm{wmed}_i\,|d_i|\Big). \tag{9}$$

A median is essential for beating partials: at the minima of a beat, where the frequency of an unresolved pair swings

furthest, the level and therefore the weight is lowest. The frequency of the partial is the weighted mean, in cents, of the f_i with $|d_i| \le \max(3\sigma_R, 1.5\,\text{cents})$, and its standard uncertainty is

$$u = \frac{\sqrt{\sigma_R^2 + 0.05\,\text{cents}^2}}{\sqrt{\max(1, D/T_N)}}, \tag{10}$$

where $D = nH/f_s$ is the duration and $T_N = N/f_s$, so that D/T_N counts non-overlapping frames. The constant term is a floor that keeps u from vanishing; it limits u to about 0.04 cents for long, stable partials. For a beating partial the deviations recur at the beat period, so that D/T_N overstates the number of independent values; u then describes the scatter of the frame estimates, and the zoom spectrum of Section 6 usually does better for a resolved pair (Section 10.2). A partial without a resolved partner may be a pair measured as a blend of its two modes. For such a partial, the shift that the zoom spectrum makes to the frequency, or half the beat rate measured from the envelope, whichever is larger, is added to u in quadrature, both in cents.

5.3. Confidence

The confidence of a partial is the product

$$\begin{aligned} c = {} & \mathrm{s}(\widetilde{\mathrm{SNR}}; 6\,\mathrm{dB}, 24\,\mathrm{dB})\left(1 - e^{-D/0.2\,\mathrm{s}}\right) \\ & \times \sqrt{P}\,\frac{1}{1 + (\sigma_R/6\,\text{cents})^2}\,\sqrt{\rho}\,\beta. \end{aligned} \tag{11}$$

P is the presence of the partial in its core interval, the shortest run of frames holding 90 % of its weight: the fraction of those frames in which it was found. ρ is the fraction of the weight carried by frames within $\max(6 \cdot 1.4826\,\mathrm{MAD}, 3\,\text{cents})$ of f_m, MAD being the unweighted median of $|d_i|$ (at least 0.3 cents). β is 1, or 0.05 for background (Section 5.4). Each factor lies in $[0, 1]$ and penalises one way of being spurious: a weak peak, a brief event, an intermittent track, a wandering frequency, a track assembled from unrelated peaks, a tone that does not belong to the instrument. A partial is listed if $c \ge \theta$ (0.7 by default), its peak level is within 60 dB of that of the strongest partial with $c \ge 0.3$, and its duration is at least $D_{\min}$ (1 s by default). The duration factor alone reaches 0.63 at 0.2 s and 0.99 at 1 s; $D_{\min}$ excludes events that are measured confidently but are too brief to matter musically. The implementation lets the user change θ, the level range and $D_{\min}$; these select which of the measured partials are listed, whereas the choice of the window and every measured quantity use the default values.

5.4. Background

A steady tone from the room or from electronics can be strong, stable and long, and pass every test above. A partial is marked as background if it was already present before the strike: a peak lies within $\max(1.5\text{ bins}, 20\,\text{cents})$ of its frequency in at least half of the frames of the 0.5 s before the attack, its level in the first 0.4 s after the attack exceeds the highest of those levels by less than 4 dB, and the slope of its level over the window is below 0.3 dB/s in magnitude. A second test, which also serves windows without an attack, takes the quietest 10 % of the frames of the signal as reference, provided they are at least 10 dB quieter than the window, and requires the median level of the partial to exceed its median level there by less than 6 dB, with the same conditions on presence and slope.

6. SPLIT MODES AND BEATING

In an axisymmetric shell each mode with nodal meridians is a degenerate pair; departures from symmetry in shape, thickness or material split the pair into two modes a fraction of a hertz to a few hertz apart, and their sum beats [1–3]. The main lobe of the analysis window spans ±23 Hz at 48 kHz, so each frame sees the pair as one peak whose frequency and level oscillate at the beat rate, and the frequency of Section 5 is a weighted average of the two modes.

6.1. Zoom spectrum

For each partial with $c \ge 0.2$ that is not background, the core interval, limited to the window and to 5 s, defines a segment of length $T \ge 0.8$ s. The segment is weighted by a Blackman–Harris window of length T, shifted to zero frequency by multiplication with $e^{-2\pi i f n/f_s}$, f being the tracked frequency, low-pass filtered and decimated by $M = \mathrm{round}(f_s/400\,\mathrm{Hz})$ with a triangular kernel spanning $2M$ samples (moving averages of lengths M and $M+1$ in cascade, with nulls near multiples of $f_s/M \approx 400$ Hz), and Fourier transformed directly on a grid of spacing $\min(0.05\,\mathrm{Hz}, 0.2/T)$ over $\pm B$,

$$B = \max\big(4/T,\ \min(12\,\mathrm{Hz},\ 0.45\,\Delta)\big), \tag{12}$$

Δ being the distance to the nearest other partial with $c \ge 0.2$ that is not background. This is the zoom transform by complex demodulation [21, 22]; evaluated on the narrow grid it costs about 10^6 operations per partial. The highest peak, refined by quadratic interpolation, replaces the tracked frequency if it lies within $\max(1\,\mathrm{Hz}, 8\,\text{cents}, 0.8\text{ bin})$ of it, the range over which an unresolved pair can pull the tracked value. A second peak at least $3/T$ from the first, no more than $\min\big(25\,\mathrm{dB}, \max(10\,\mathrm{dB}, \widetilde{\mathrm{SNR}} - 10\,\mathrm{dB})\big)$ below it and not within two grid steps of $\pm B$ is reported as the partner mode; the beat rate is the distance between the two. A resolved pair accounts for the frame-to-frame wobble of the frequency, and the stability factor of (11) is then raised to at least 0.95.

6.2. Envelope beat

A pair that decays within about a second cannot be resolved, since the zoom window's main lobe spans $\pm 4/T$, but it still modulates the level. Over the whole track of the partial (at least 0.6 s; the core interval of a fast-decaying partial is too short), its level is interpolated across gaps and detrended by a straight line in dB, i.e. by an exponential decay. If the residual r_j spans at least 2 dB between its 5th and 95th percentiles, its normalised autocorrelation

$$\gamma(L) = \frac{n}{n-L}\,\frac{\sum_j r_j r_{j+L}}{\sum_j r_j^2} \tag{13}$$

is searched for its highest local maximum between lags of 0.15 s and $\min(2.5\,\mathrm{s}, nH/2f_s)$. If that maximum reaches 0.5 and the track spans at least two of its lags, the beat rate is the inverse of the lag, refined by parabolic interpolation, and the uncertainty of the partial is widened as described in Section 5.

7. DISPLAY AND AUDIBILITY

7.1. Spiral

As in the Snail [4], the spectrum is drawn on a spiral with one turn per octave, the pitch helix [23, 24] seen along its axis. A frequency f is placed at

$$u = \log_2(f/f_{\mathrm{lo}}), \quad \theta = 2\pi u, \quad \rho = r_0 + \Delta r\, u, \tag{14}$$

with the angle θ measured clockwise from the top and f_{lo} the C at the bottom of the displayed range (C2, over seven octaves, by default). The C line points upwards, and the twelve pitch classes of the equal-tempered scale lie on twelve radial lines.

7.2. Reassigned spectrum

For the window, the power of each bin is moved to its reassigned frequency (6) and accumulated on a fixed grid of 2-cent cells from C1 to C10: bin k of frame ℓ adds $4|X_h[\ell,k]|^2/(N\sum_n h[n]^2)$, spread linearly over the cells within 4 cents of $\hat{f}[\ell,k]$, provided $|\hat{f}[\ell,k] - kf_s/N| \le 3.5$ bins, i.e. the estimate lies inside the main lobe. The factor makes the cells of a stationary sinusoid of amplitude A sum to A^2. This is the reassigned spectrogram of [11, 12], reduced to its frequency coordinate. Each contribution is weighted by the level of the bin above the local noise floor and by its lobe coherence, as in the real-time display (Eq. (19), Section 9), so that noise and transients leave the spiral dark; the measurement does not use these weights. The cells are averaged over the frames of the window (or their maximum is taken, as an option), and the cell of each listed partial is raised to the partial's peak level, so that partials that die out early in the window keep their height. Cell levels are mapped to $v_c \in [0,1]$ over a display range below the maximum (60 dB), widened to lenses,

$$v'_c = \max_j \left(v_{c+j} - (2j/W)^2\right), \tag{15}$$

of half-width $W\sqrt{v}$ cents ($W = 22$ cents), and drawn as a band of half-thickness $0.47\,\Delta r\, v'_c$ across the spiral, coloured by v'_c (Fig. 1).

7.3. Audibility

The level of a partial says little about how prominently it is heard. Each listed partial is weighted by the equal-loudness contour at 60 phon of ISO 226:2003 [25],

$$w_p = L_p - \left(L_{60}(f_p) - 60\,\mathrm{dB}\right), \tag{16}$$

L_p being its peak level relative to the strongest partial and $L_{60}(f)$ the sound pressure level of the contour, computed from the standard's formula at its 29 frequencies and interpolated in $\log f$. Relative to the largest w_p, a partial is *dominant* above -10 dB, *clear* above -25 dB, *faint* above -40 dB and *very faint* below. It is moved one class down if a partial at least 15 dB stronger in w lies within one unit of the ERB-number scale [26],

$$E(f) = 21.4 \log_{10}(1 + 0.00437 f), \tag{17}$$

as a crude allowance for masking. The calibration of a recording is unknown, so 60 phon stands for a moderate listening level; the classes are a guide to prominence, not a loudness model.

8. MICROPHONE INPUT

The microphone signal reaches the analysis through an audio worklet in blocks of 1024 samples and is kept in a ring buffer of 20 s. Frames are analysed as they arrive; while the microphone is on, the spiral shows the real-time display of Section 9. An attack opens a capture, which is closed by the next attack of comparable strength, after $T_{\max}$, or, once it is 1 s old, when no confident partial has been seen for 0.6 s. While no capture is open and confident partials are sounding, a capture of length $T_{\max}$ is also taken every 2.5 s, for sustained sounds; unlike the windows of Section 4, these are not required to be free of attacks. Each capture is analysed as in Sections 4–6, with the quietest frames of the last 30 s as background reference and the zoom spectra computed from the ring buffer, and its end is trimmed as in Section 4 without a second analysis. Captures without a listed partial are discarded. The best capture by (8) is shown; once there are two or more, each partial is reported with the number of captures in which it, or the other mode of its pair, was found within 15 cents. In the real-time mode of Section 9 the microphone feeds only the real-time display.

9. REAL-TIME DISPLAY

The measurement reports partials once a ring-out has been analysed. For watching sound as it happens, in particular when several instruments sound at once and attacks carry no meaning, the implementation has a second mode, *Observe*, with a display of its own. It is built, like the rest of the method, from quantities of single frames; only levels are carried from one frame to the next.

9.1. Frames and lobe coherence

Frames of $N = 8192$, 4096 or 2048 samples (171, 85 or 43 ms at 48 kHz; 8192 by default) are taken every $H = 256$ samples (5.3 ms) with the window (3). Besides X_h and $X_{\mathscr{D}h}$, two further FFTs are computed with the time-weighted windows $\mathscr{T}h[n] = (n - N/2)\,h[n]$ and $\mathscr{T}\mathscr{D}h[n] = (n - N/2)\,\mathscr{D}h[n]$. The lobe-coherence test of Section 3 compares the reassigned frequencies of neighbouring bins; its continuous counterpart is the derivative of the reassigned frequency with respect to the analysis frequency. Differentiating (4) with $\partial X_g/\partial\omega = -iX_{\mathscr{T}g}$ (the terms in the time origin cancel) gives

$$\mu = \frac{\partial\hat{\omega}}{\partial\omega} = 1 + \mathrm{Re}\left\{\frac{X_{\mathscr{T}\mathscr{D}h}}{X_h} - \frac{X_{\mathscr{D}h}X_{\mathscr{T}h}}{X_h^2}\right\}, \tag{18}$$

which is one plus the mixed partial derivative $\partial^2\Phi/\partial t\,\partial\omega$ of the STFT phase in the form given by Auger, Chassande-Mottin and Flandrin [30]. For a stationary sinusoid every bin of the main lobe reassigns to the same frequency and $\mu = 0$; for an impulse no bin is moved and $\mu = 1$; Nelson used these two values to tell line components from impulses [27–29]. For a linear chirp μ lies between them: with a Gaussian window of time spread λ and a chirp rate α (rad/s^2), $\mu = \alpha^2\lambda^4/(1+\alpha^2\lambda^4)$ [30]. In noise μ scatters widely: in white noise, half the bins have $|\mu| > 0.76$.

Table 1: Recorded bowl, window 18.62–23.80 s, default settings ($\theta = 0.7$, $D_{\min} = 1$ s, $f_{A4} = 440$ Hz), as exported by the implementation. Level: peak level relative to the strongest partial; u: standard uncertainty (10), widened for #4 as described in Section 5.

#	note	frequency (Hz)	ratio	level (dB)	audibility	confidence	u (cents)	beat (Hz)	partner (Hz)
1	A#3+42	238.87	1.000	−3.3	clear	0.99	0.14	1.11	237.76
2	D#5-11	618.46	2.589	0.0	dominant	0.99	0.10	0.95	619.42
3	D6-47	1143.27	4.786	−3.2	dominant	1.00	0.06	0.84	1142.42
4	A6+38	1799.24	7.532	−14.4	clear	0.99	1.45	3.01	–

9.2. Deposits

As in Section 7, the power of each bin is moved to its reassigned frequency (6) on the grid of 2-cent cells, if that frequency lies inside the main lobe, but with two weights:

$$e[\ell,k] = \frac{4|X_h[\ell,k]|^2}{N\sum_n h[n]^2}\, e^{-(\mu[\ell,k]/0.3)^2}\, \mathrm{s}\big(S[\ell,k]; 3\,\mathrm{dB}, 15\,\mathrm{dB}\big). \tag{19}$$

S is the level of the bin above the local noise floor, the median power of its one-third-octave band (Section 3) divided by $\ln 2$, the ratio of the median to the mean of an exponential variate. The level weight removes most of the noise, and also broadband events such as taps, which raise the band median with them. The coherence weight removes the remaining noise and the residue of transients, and also the energy that the window smears across many bins during a fast glide; with $N = 4096$ it falls to 0.5 at a chirp rate of about 600 Hz/s (about 2400 and 150 Hz/s for $N = 2048$ and 8192), which a vibrato of $\pm 0.8\,\%$ at 5.5 Hz reaches above 2 kHz. Each deposit is spread linearly over the cells within 4 cents of its frequency.

9.3. Smoothing in decibels

Let $E_c[\ell]$ be the sum of the deposits in cell c in frame ℓ, $L_{\max}[\ell]$ the largest of them on the displayed part of the spiral, held and falling by at most 3 dB/s, and $L_{\mathrm{fl}} = L_{\max} - R - 25\,\mathrm{dB}$, R being the display range. The displayed level of the cell follows

$$\Lambda_c[\ell] = \Lambda_c[\ell-1] + \alpha\left(\max(10\log_{10} E_c[\ell], L_{\mathrm{fl}}) - \Lambda_c[\ell-1]\right), \tag{20}$$

with $\alpha = 1 - e^{-H/(f_s\tau)}$, $\tau = 40$ ms while the level rises and 170 ms while it falls by default (*steady*; 15 and 60 ms for the option *medium*, 8 and 30 ms for *fast*): the attack and release of a level meter, applied to each cell. Averaged in decibels, a cell that receives energy only in occasional frames, as from residual noise, stays near the floor, whereas a partial that deposits its energy frame after frame rises to its level within a few time constants. Λ is mapped to $[0,1]$ over the range R below $L_{\max}$ and drawn as in Section 7. For a recording, the display at any moment is obtained by analysing the 0.3 s before it, and during playback it follows the sound.

9.4. Read-out

The display follows the sound within a few tenths of a second, faster than a list of numbers can be read. The list of partials is therefore built from read-outs of the display taken four times a second and is renewed once a second. A read-out takes the maxima of Λ on the displayed part of the spiral that exceed every cell within 12 cents and lie within 46 dB of the strongest. The frequency of each is the centroid of the reassigned frequencies deposited in the nine cells around it, weighted by the deposits (19) and smoothed with the release time constant; its signal-to-noise ratio is the mean of S over the same deposits. Maxima are followed from one read-out to the next by frequency, within 25 cents, and two followed partials that come within 15 cents of each other are merged. A partial joins the list once it has been found in three of the last four read-outs with a median level within 40 dB of the strongest maximum, or of −50 dB (re full scale) if that is stronger, and a median signal-to-noise ratio of at least 12 dB. The last two conditions keep background noise out of the list when nothing else sounds. A listed partial is shown dimmed after three read-outs without it and leaves the list after six, or when it falls 6 dB below either limit. The frequency and level shown are the medians over the last 1.5 s. At most 16 partials are shown, the strongest, those already on show counting 3 dB stronger so that partials of nearly equal level do not take turns, and dimmed ones 6 dB weaker so that they give way to partials that sound. The audibility class is computed as in Section 7. The list states what has sounded steadily during the last second or so. It carries neither a confidence nor an uncertainty, which only the measurement of Sections 4–6 provides.

10. EVALUATION

10.1. Single-frame accuracy

A sinusoid of random phase, with frequency uniform over one bin above 1 kHz, in white Gaussian noise, was analysed in one frame ($f_s = 48$ kHz, $N = 8192$, 300 trials per SNR, the SNR per sample being $A^2/2\sigma^2$ for amplitude A and noise variance σ^2). Fig. 3 compares the root-mean-square error of the reassigned frequency (6) at the highest bin with that of quadratic interpolation of the log magnitude [8, 18], and with the Cramér–Rao bound for a sinusoid observed over N samples [31]. The reassigned frequency stays within a factor 2.1 to 2.6 of the bound from −20 to 45 dB SNR per sample (the processing gain of the frame, by which the ratio of the sinusoid's peak power to the noise power in a bin exceeds the SNR per sample, is $N/(2B_e) = 33$ dB, $B_e = 2.00$ bins being the equivalent noise bandwidth of the window [14]); the factor is the price of the window's taper. The interpolated estimate levels off at 0.023 cents (at most 0.032 cents), the bias of the parabolic fit [18]. Above 45 dB the reassigned estimate levels off at about $4\cdot10^{-4}$ cents RMS ($2.5\cdot10^{-4}$ Hz), the limit set by the window's end values. With a decay time constant of 2 s the noise-free error remains below $7\cdot10^{-4}$ cents. Both estimators are far more precise than the tuning of a bowl requires; what limits the measurement in

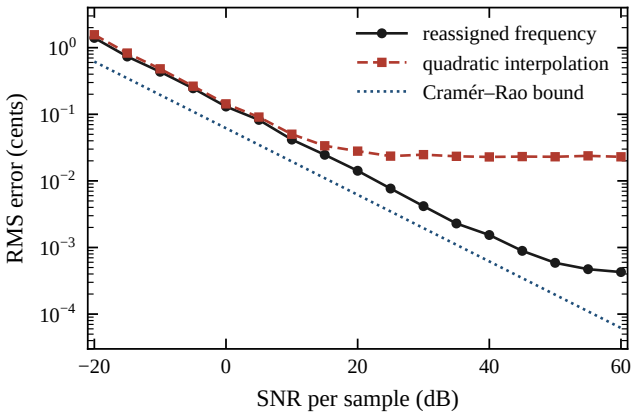


Figure 3: Single-frame frequency error for a sinusoid near 1 kHz in white noise ($N = 8192$, $f_s = 48$ kHz; 1 cent is 0.58 Hz here), against the Cramér–Rao bound [31].

practice is the behaviour of the partials themselves, chiefly beating.

10.2. Synthetic ring-outs

Two struck bowls were synthesised as in (1), each with a 3 ms noise burst for the strike at $t_0 = 3$ s. Table 2 gives the results with the default settings.

Case A (44.1 kHz) has seven partials with time constants from 12.5 to 1.7 s, among them two split pairs, 280.20/283.90 Hz (amplitudes 0.05/0.03) and 2314.00/2317.50 Hz (0.02/0.01), separated by about 17 and 16 times $1/T$; a steady background tone at 11926.8 Hz of amplitude $4 \cdot 10^{-4}$, present throughout; and white noise of standard deviation $3.5 \cdot 10^{-5}$, so that the median SNRs of the partials lie between 69 and 91 dB. The attack is detected in the block containing t_0 and placed 8 ms before it, and the window 3.01–8.20 s is chosen. The frequencies of the five partials and of both partner modes are recovered to within $8 \cdot 10^{-6}$ Hz, the beat rates to $3 \cdot 10^{-6}$ Hz, and the partner levels as synthesised, $20\log_{10} 0.6 = -4.4$ dB and $20\log_{10} 0.5 = -6.0$ dB. Before refinement the lower pair had been tracked at 280.84 Hz, between its modes. The background tone is detected and its confidence reduced to 0.05.

Case B (48 kHz) takes parameters close to those of the recorded bowl of Section 10.3: four pairs with its frequencies and partner ratios, decaying at 1.1, 0.8, 4.8 and 27 dB/s; the last has nearly equal modes 2.98 Hz apart (1799.24/1802.22 Hz, amplitude ratio 0.69), in white Gaussian noise of standard deviation 0.003 (median SNRs from 25 to 56 dB). The three slower pairs are resolved, although their modes are only 3 to 5 times $1/T$ apart: the main modes are found within 0.0003, 0.0000 and 0.023 cents of the truth and the partners within 1.3, 0.2 and 40 mHz, the last pair, at about $3/T$, being the hardest. Their uncertainties, 0.19, 0.10 and 0.06 cents, computed from the frame-to-frame wobble, cover these errors. The fast pair is not resolved. It is reported at 1800.24 Hz, between its modes and 0.97 cents above the stronger, with its beat of 2.98 Hz from the envelope and an uncertainty of 1.44 cents. At half the noise the errors of the three resolved main modes fall below 0.001 cents, and without noise all four pairs are resolved, to $3 \cdot 10^{-4}$ cents.

Table 2: Synthetic ring-outs: synthesised and measured frequencies (Hz), confidence c and uncertainty u (cents). Partner modes are given below the main mode of each pair.

synthesised	measured	c	u	
Case A				
280.20	280.200 00	0.94	0.84	beat 3.700
283.90	283.900 00			partner, −4.4 dB
802.93	802.930 00	1.00	0.04	
1497.21	1497.210 00	1.00	0.04	
2314.00	2314.000 00	0.99	0.11	beat 3.500
2317.50	2317.500 00			partner, −6.0 dB
3277.39	3277.389 99	1.00	0.04	
11926.80	11926.800 40	0.05	0.04	background
Case B				
238.87	238.870 0	0.97	0.19	beat 1.109
237.76	237.761 3			partner, −17.9 dB
618.46	618.460 0	0.99	0.10	beat 0.960
619.42	619.419 8			partner, −14.5 dB
1143.27	1143.285 2	1.00	0.06	beat 0.905
1142.42	1142.380 1			partner, −5.4 dB
1799.24	1800.243 4	0.99	1.44	beat 2.978 (envelope)
1802.22				partner, unresolved

10.3. A recorded bowl

Figs. 1, 4 and 5 and Table 1 show the analysis of a 47 s recording of a singing bowl at 48 kHz, its two channels averaged. Two attacks are detected (Fig. 4a): a light one at 2.0 s and a strike at 18.6 s, 32 dB stronger; between 5 and 11 s the level rises without an attack. The ring-out window 18.62–23.80 s scores 4.37; the best sliding window, 20.05–25.23 s later in the same ring-out, scores 2.95, and the best over the passage without a strike, 7.52–12.69 s, 2.80. Of the 109 partials formed in the window, four are listed, at frequency ratios 1 : 2.589 : 4.786 : 7.532. Among the others, a partial at 2577.68 Hz has confidence 0.82 but lasts 0.35 s and is excluded by $D_{\min}$; none of the rest exceeds 0.39.

The three lower partials are split pairs (Fig. 5). Their partners lie 1.11, 0.95 and 0.84 Hz away and 17.9, 14.5 and 5.2 dB below the main mode. For an amplitude ratio r the level of a pair swings by $20\log_{10}\big((1+r)/(1-r)\big)$, here 2.2, 3.3 and 10.7 dB, close to the swings of the detrended level curves of Fig. 4b, 2.9, 4.0 and 10.3 dB between their 5th and 95th percentiles. The partial at 1799.24 Hz decays at 23 dB/s, against 0.8 to 4.8 dB/s for the others. Its zoom segment of 1.0 s cannot separate modes 3 Hz apart, and its beat of 3.01 Hz is measured from the envelope; its level swings by 14 dB, which implies nearly equal modes (amplitude ratio about 0.67), the situation of the fast pair of Case B. Its uncertainty, 1.45 cents, therefore includes half the beat rate. When the recording is played into the microphone path of Section 8, the capture of this ring-out gives a zoom segment of 1.14 s for this partial, and the pair is resolved there, at 1799.22 and 1802.17 Hz, 0.8 dB apart. Refinement moved the tracked frequencies by +0.02, +0.01, +0.11 and −1.11 Hz.

10.4. Real-time display

Table 3 compares the display of Section 9 with the unweighted reassigned spectrum of Section 7 shown frame by frame, the cell powers of 8192-point frames averaged

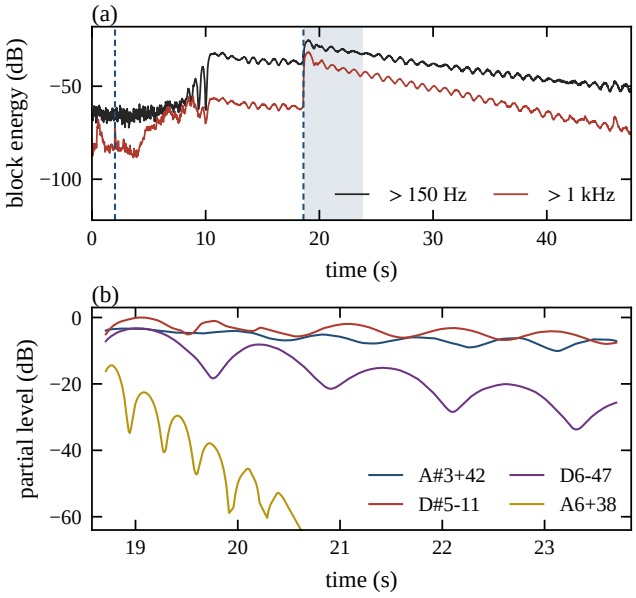


Figure 4: Recorded bowl. (a) Block energies of the onset detector over the whole recording; dashed: detected attacks; shaded: the selected window. (b) Peak levels of the four listed partials in the window, relative to the strongest; the periodic dips are beats.

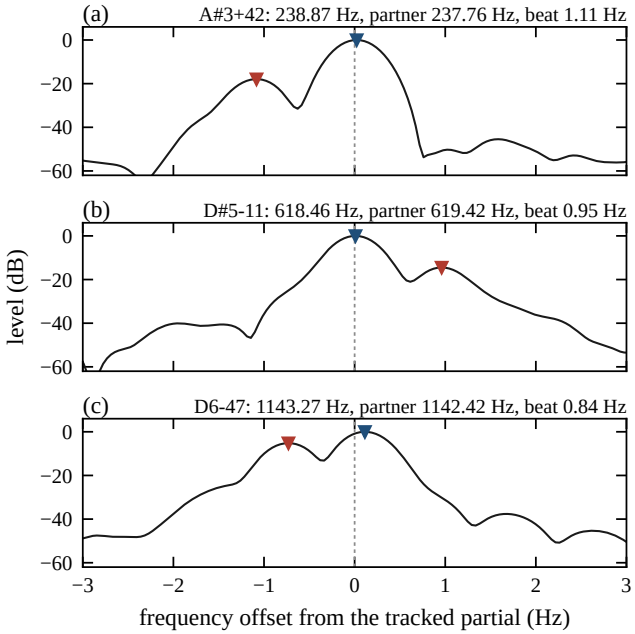


Figure 5: Zoom spectra of the three lower partials of the recorded bowl over their core intervals ($T = 4.5$ to 4.7 s). Blue: main mode, red: partner; dashed: the frequency tracked from the frames.

with weight 0.4 for the newest frame, and with variants of itself. All signals are synthetic and sampled at 48 kHz. Six steady partials, at 196 to 3413 Hz and 0 to −34 dB, in white noise of standard deviation 0.05 or 0.2 (per-bin signal-to-noise ratios of the weakest partial, for $N = 4096$, of about 19 and 7 dB), give the fraction of the spiral away from the partials that is visibly lit, more than 3 % of the display range, and the flicker of the weakest partial, the standard deviation of its displayed level. A tone at 660 Hz, switched on and off beside one at 440 Hz, gives the time until it is shown within 6 dB of its steady level and the time until it has fallen to a tenth of it on the display, from the moment the change reaches the newest sample of a frame (on a 20 ms grid). A scene of 6 s combines a struck bowl, its four partials split as in Table 1 and decaying, a sung melody of five notes with a vibrato of ±0.8 % at 5.5 Hz and glides of 60 ms between the notes, four taps (bursts of noise decaying with 2 ms), and pink noise. The extent of the spiral that is lit, in cents, is averaged over quiet stretches and over the 150 ms after each tap and after the start of each glide (Fig. 6).

The level weight keeps noise off the display in both noise conditions and removes the taps. The coherence weight then reduces the extent lit in the scene by a third, and during glides by three fifths, so that it hardly changes with taps and glides; compared with the unweighted power average, the extent lit falls by 85 % after taps and by 80 % during glides. At equal frame length the smoothing in decibels removes a stopped partial in about half the time of the power average (160 against 304 ms for $N = 8192$). The price of the level weight is the flicker of a partial only a few decibels above the noise floor of a frame: 7 dB for the weakest partial in the stronger noise, 4.6 dB with $N = 8192$, where its signal-to-noise ratio is 3 dB higher. Such a partial stays below the 12 dB required for the list. The listed frequencies of the five stronger partials have RMS errors of 0.03 to 0.13 cents in the stronger noise. With $N = 8192$ and the steady smoothing, the setting with which the implementation starts, the flicker of the weakest partial is 3.5 dB and these errors are 0.02 to 0.06 cents, while a partial takes 200 ms to be shown in full and 261 ms to disappear. Read four times a second, the maxima of the display enter and leave by 37 partials per second in the melody of the scene and by 11 per second when the voice holds a single note; the list renewed every second changes by 14 partials per second in the melody, nearly all at the changes of note, and by 0.3 per second for the held note, with 14 to 16 rows. In white noise alone, at −60 dB re full scale, the maxima of the display enter and leave by 140 per second, and the list stays empty. On the recorded bowl, the listed frequencies of the three lower partials, renewed every second from 19.5 to 23.5 s, agree with the measured ones of Table 1 to within 0.4 cents in the median and vary by at most 0.5 cents from one list to the next.

10.5. Computation

A frame, two FFTs of 8192 points and the peak analysis, takes 0.7 to 0.8 ms in JavaScript (V8, Node.js 22) on one core of a cloud virtual machine, under 8 % of the hop of 10.7 ms; the 47 s recording, 4430 frames, is analysed with all candidate windows in about 5 s. A frame of the real-time display, four FFTs of 4096 points and the update of the cells, takes 0.5 ms, under 10 % of its hop of 5.3 ms.

11. RELATION TO THE SNAIL-ANALYSER

Both methods estimate, for each analysis frequency, the frequency of the component in its spectral lobe, and use the estimate to sharpen a spiral display. They differ in how the estimate is formed, in what sharpens the display, and in what they deliver.

The Snail interpolates the STFT onto a logarithmic grid of analysis frequencies, typically 20 per semitone, with frames

Table 3: Real-time display, synthetic signals. Lit: share of the spiral away from the partials shown above 3 % of the display range; flicker: standard deviation of the displayed level of the weakest partial (−34 dB); full, gone: time until a partial switched on is shown within 6 dB of its level, and until one switched off has fallen to a tenth of its display value; scene: extent of the spiral lit in the scene of Fig. 6; read-out: RMS error of the frequency in the list, renewed every second from 2 to 8 s, of the partials at 541.7 Hz (−6 dB) and 3412.8 Hz (−34 dB) in the stronger noise; –: not listed. Observe: $N = 4096$, $\tau = 15/60$ ms; the rows below it change the frame length or the smoothing, the last row both, to the setting with which the implementation starts.

	lit (%)		flicker (dB)		full	gone	scene, lit (cents)			read-out (cents)	
display	σ=0.05	0.2	0.05	0.2	(ms)	(ms)	quiet	taps	glides	−6 dB	−34 dB
unweighted power average	4.0	33.1	0.4	2.3	120	304	477	2079	1219	–	–
Observe without weights	4.8	24.0	0.4	2.5	120	144	405	1538	1186	0.07	0.39
level weight only	0.0	0.3	0.4	7.1	120	123	379	478	590	0.07	–
Observe	0.0	0.1	0.5	7.1	120	123	257	316	241	0.07	–
fast smoothing	0.0	0.3	0.6	7.5	83	80	230	280	263	0.14	–
steady smoothing	0.0	0.0	0.3	6.0	163	203	320	413	320	0.03	–
$N = 2048$	0.0	0.0	0.6	3.8	83	101	252	306	215	0.04	–
$N = 8192$	0.0	0.5	0.4	4.6	163	160	215	294	235	0.03	–
$N = 8192$, steady smoothing	0.0	0.2	0.3	3.5	200	261	246	339	287	0.02	–

of 50 ms every 5 ms [5]. Its frequency deviation is the change of the demodulated phase $\varphi_d(t,f) = \varphi(t,f) - 2\pi ft$ between successive frames, divided by 2π times the hop, and its display weights the loudness at each analysis frequency by the modulus of the low-pass filtered phasor $e^{i\varphi_d}$; this weighting contracts the lobe of a stationary component to a width set by the cut-off frequency and suppresses components that vary faster [5, 6]. The present method forms neither the demodulated phase nor a phase difference between frames. Its frequency estimate is the reassigned frequency (6) of a single frame, on the linear grid of the FFT, and its display is sharpened by moving power to that frequency, without a weighting by temporal consistency. As the hop tends to zero the two frequency estimates tend to the same quantity, the instantaneous frequency of the band-pass signal of the bin [11,12]: the Snail evaluates it by a finite difference across frames, the present method analytically within one frame.

The measurement trades reactivity for statements of accuracy. The Snail follows vibrato and glissandi within tens of milliseconds, presents a clean image in noise through its constancy weighting, and adds tuning and harmonic-synchronicity indicators for performers [5]. The measurement uses frames more than three times longer and reports partials only once a ring-out of at least $D_{\min}$ has been analysed; noise is kept out of its list by the tests of Sections 3 and 5, and out of its display by the weights of the real-time display. In return it chooses its analysis window, gives each partial a confidence and a standard uncertainty, and resolves split modes a fraction of a hertz apart, which requires an observation of several seconds (Section 6); none of these is part of the published descriptions of the Snail [4, 5].

The real-time display of Section 9 serves the Snail's purpose, a clean, stable and reactive image of several sounds at once, by other means. It forms no demodulated phase, no phase difference between frames and no low-pass filtered phase quantity. Its image is sharpened by moving power to the reassigned frequency rather than by weighting the loudness at fixed analysis frequencies, and its weights are computed within one frame: the level of a bin above its band's median, and the derivative (18) of the reassigned frequency across frequency, which states whether the bins of a lobe agree on one frequency, not whether a phase stays constant in time. What is carried from frame to frame is the level of each cell, smoothed in decibels as by a level meter. The Snail's constancy indicator suppresses components that vary faster than its cut-off frequency [5]; the coherence weight similarly dims components that sweep faster than about 150 Hz/s with the default frames (Section 9), so that the upper harmonics of a strong vibrato fade during the fastest part of each cycle, and the shorter frames reduce the effect.

12. CONCLUSION

The method measures the sustained partials of a struck bowl on the ring-out it selects itself, and states how far each measurement can be trusted. Its frequency estimates come from reassignment within single frames, which in continuous time is exact for an exponentially damped complex exponential, and which in noise stays within a factor of 2.1 to 2.6 of the Cramér–Rao bound; at that precision the measurement is governed by the instrument. Split modes that last long enough are resolved by a zoom spectrum over the ring-out, to within 0.03 cents with the parameters of the recorded bowl; a pair that decays too fast is reported as a blend of its modes, with its beat rate and a correspondingly wider uncertainty. The confidence (11) and the minimum duration remove brief and unstable events, and background tones are recognised by their presence before the strike. The spiral display shows the reassigned spectrum from which the partials were measured, so that what the list reports can be seen. For observing several instruments in real time, a second display weights the reassigned power of each frame by its level above the noise floor and by the agreement of its lobe, and smooths it in decibels; it keeps noise and transients off the spiral and follows the sound within a few tenths of a second, again without carrying phase from one frame to the next. The implementation, a single HTML file, is released under the GNU General Public License, version 3 or later.

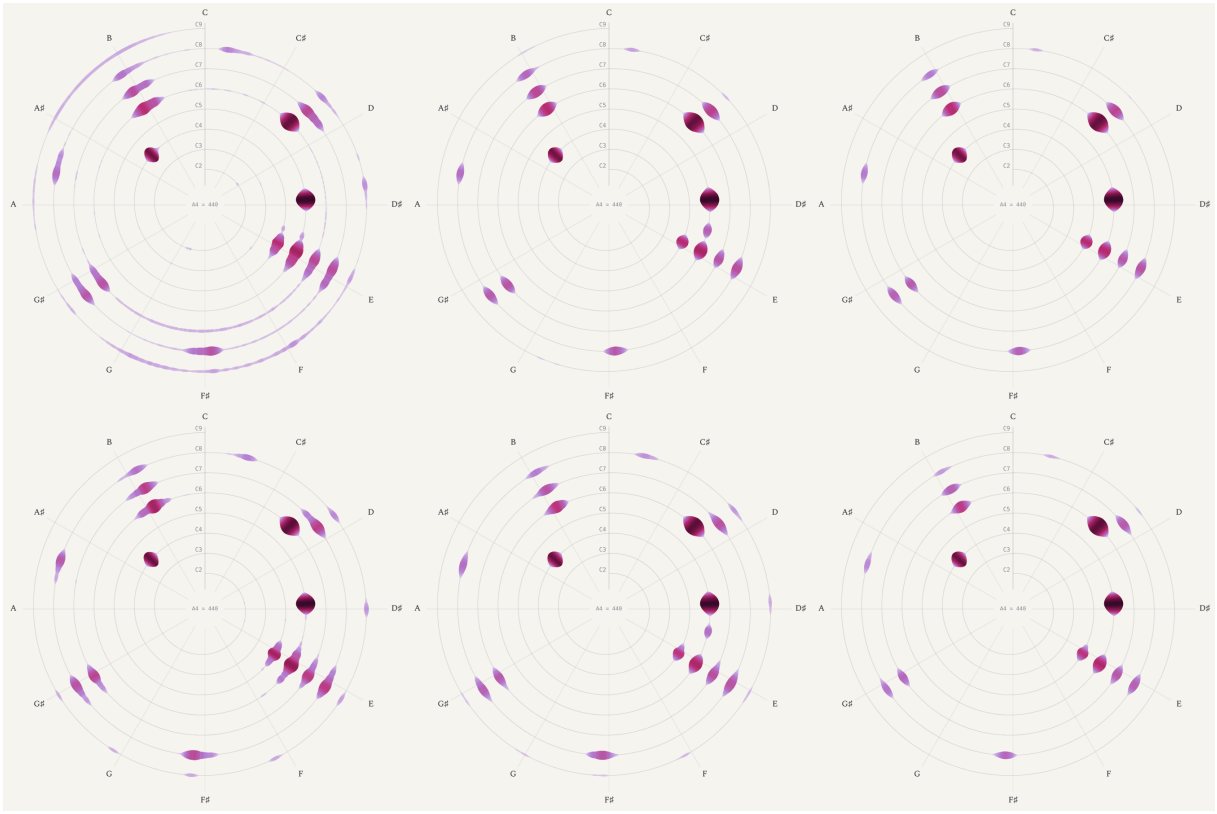


Figure 6: The scene of Section 10.4 at 2.34 s, 40 ms after a tap (top), and at 2.64 s, 40 ms into the glide from E4 to G4 (bottom). Left: unweighted reassigned power, averaged frame by frame; centre: real-time display with the level weight only; right: real-time display (19)–(20). Display range 60 dB; the bowl's partials are at A#3, D#5, D6 and A6, the melody's harmonics elsewhere.

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